

WASP-50b

R-1180

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_151226 · created 2026-07-31T04:31:36+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457382.732 +/- 0.051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.05
Transit depth (Rp/Rs)^2	3.01 +/- 1.74 [%]
Orbital Inclination (inc)	89.63 +/- 2.91
Airmass coefficient 1 (a1)	0.121 +/- 0.065
Airmass coefficient 2 (a2)	1.26 +/- 0.61
Scatter in the residuals of the lightcurve fit is	40.6 %
Best Comparison Star	None
Optimal Aperture	2.83
Optimal Annulus	13.43
Transit Duration (day)	0.071 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.05 vs 0.1404 (deviation 0.65σ)
Duration fit vs published	0.071 d vs 0.0754 d (deviation 0.16σ)
Mid-time O–C	25.4 min
Depth SNR	3.5
Residual scatter	40.6 %

Light curve

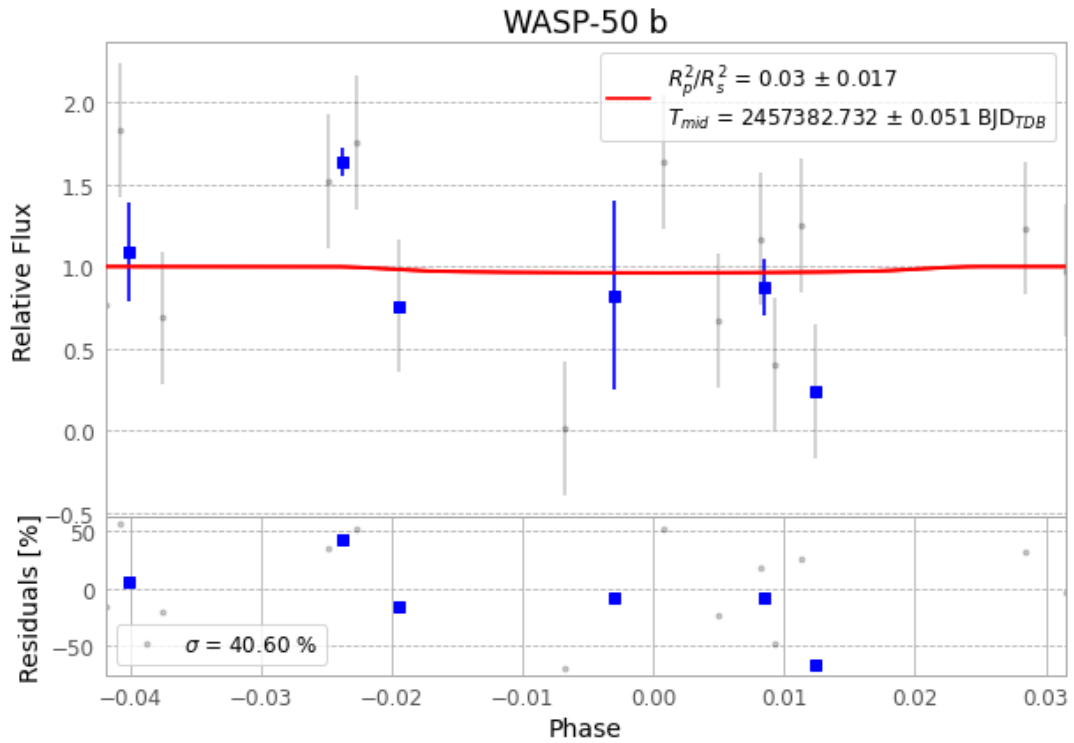


Figure 1 — FinalLightCurve_WASP-50 b_2015-12-25.png (SHA-256 a6556cbbbcd58bf0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [443, 145], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ac9c2ae32d8ee8b0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-50151226032112.FITS → WASP-50151226070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-151226.log (44c5cde4ad0e...), FinalLightCurve_WASP-50 b_2015-12-25.png (a6556cbbbcd5...), FinalLightCurve_WASP-50 b_2015-12-25.csv (7d6e77facebf...)

Compute resources

Wall time 315.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 04:31Z.